

Linear algebra: Exercise session 4

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1. Assume that $A \in \mathbb{R}^{m \times n}$. Show that the following four fundamental sets are subspaces:

$$C(A) = \{Ax : x \in \mathbb{R}^n\}, \quad C(A^T) = \{A^T x : x \in \mathbb{R}^m\}, \quad N(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \quad N(A^T) = \{x \in \mathbb{R}^m : A^T x = 0\}.$$

2. Find bases for four fundamental spaces of the following matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 2 & 4 \\ 3 & 8 & 6 & 16 \end{bmatrix}, \quad C = \begin{bmatrix} -1 & 2 \\ 1 & -2 \\ 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 2 & 1 & 5 \\ 2 & 4 & -3 & 0 \\ 1 & 2 & -1 & 1 \end{bmatrix}.$$

3. Let $A \in \mathbb{R}^{m \times n}$ with $m \leq n$ and system $Ax = b$ is solvable for every $b \in \mathbb{R}^n$. Describe the column and left nullspace and calculate the dimension of four fundamental spaces.
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4. Let $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$ and the system $Ax = b$ has a unique solution. Describe the column, row and (left) nullspaces and calculate the dimension of four fundamental spaces.
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5. Construct a 3×3 matrix A whose column space contains $(1, 1, 0)^T$ and $(1, 0, 1)^T$ but not $(1, 1, 1)^T$.
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6. Construct a 3×3 matrix A whose column space is only a line.
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7. Let A is a 7×9 matrix with $\text{rank}(A) = 5$. What are the dimensions of the four subspaces? What is the sum of all four dimensions?
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8. Let A is a 3×4 matrix with $\text{rank}(A) = 3$. What are its column space and left nullspace and the dimension of four fundamental subspaces?
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9. Does the set of all 3×3 matrices of rank 3 forms a linear space? How about the set of all 3×3 matrices of rank less than or equal to 2?
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